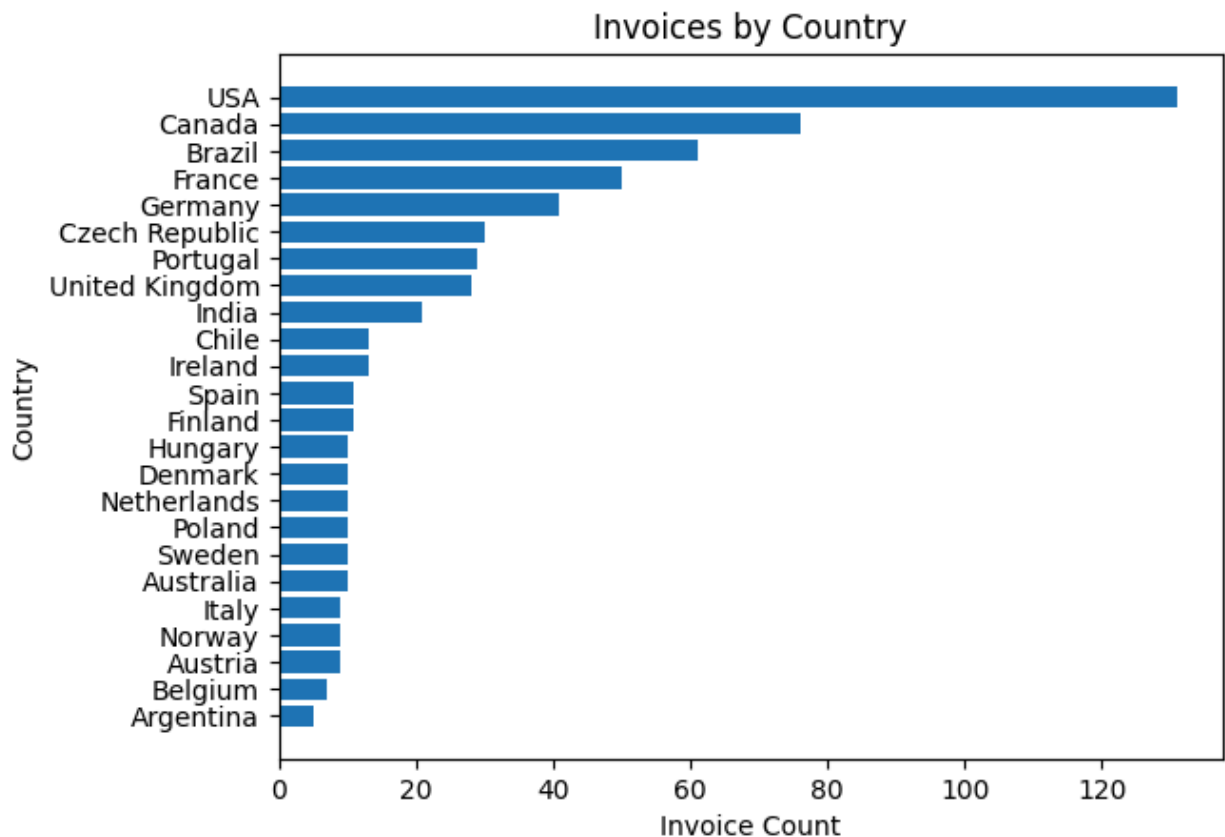


#1. Who is the senior most employee based on job title?

```
select *  
FROM Musicstore_pro.employee  
order by levels desc  
limit 1;
```

#2. Which countries have the most Invoices?

```
select billing_country, count(*) as count_invo  
from Musicstore_pro.invoice  
group by Billing_Country  
order by count_invo desc;
```



Key Insights

- **USA dominates** with the highest invoices (131) — far ahead of others

- **Canada & Brazil** are strong secondary markets
- European countries are **moderately distributed** (France, Germany, UK, etc.)
- **India is mid-tier** with 21 invoices
- Many countries cluster around **~10 invoices**, showing a long tail

#3. What are top 3 values of total invoice?

```
select invoice_id, round(sum(total), 2) as total_sum
from Musicstore_pro.invoice
group by invoice_id
order by total_sum desc
limit 3;
```

Row	invoice_id	total_sum
1	183	23.76
2	362	19.80
3	92	19.80

Quick Insight

- Invoice **183** has the highest total (23.76)
- Invoice **362** and **92** are tied

/*4. Which city has the best customers? We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals

*/

```
select billing_city,
       sum(total) as total
from Musicstore_pro.invoice
group by billing_city
order by total desc
limit 1;
```

Row	billing_city	total
1	Prague	273.24

Quick Insight

- **Prague** has the **highest total invoice amount**

/*5.Who is the best customer? The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money.*/

```
select c.first_name,
       c.last_name,
       round(sum(i.total),2) as total_spend
from Musicstore_pro.customer c
join Musicstore_pro.invoice as i on c.customer_id = i.customer_id
group by c.first_name,c.last_name
order by total_spend desc
limit 1;
```

Row	first_name	last_name	total_spend
1	František	Wichterlová	144.54

Quick Insight

- **František Wichterlová** is the **highest spending customer**
- Total spend: **144.54**

/*6. Write query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A.*/

```
select c.email,
       c.first_name,
       c.last_name
from Musicstore_pro.customer c
join Musicstore_pro.invoice i on c.customer_id = i.customer_id
join Musicstore_pro.invoice_line l on i.invoice_id = l.invoice_id
where l.track_id in (
    select track_id
    from Musicstore_pro.track t
    join Musicstore_pro.genre g on t.genre_id = g.genre_id
    where g.name like "Rock"
)
order by c.email ;
```

Output:

Row	email	first_name	last_name
1	aaronmitchell@yahoo.ca	Aaron	Mitchell
2	alero@uol.com.br	Alexandre	Rocha
3	astrid.gruber@apple.at	Astrid	Gruber
4	bjorn.hansen@yahoo.no	Bjørn	Hansen
5	camille.bernard@yahoo.fr	Camille	Bernard
6	daan_peeters@apple.be	Daan	Peeters
7	diego.gutierrez@yahoo.ar	Diego	Gutiérrez
8	dmiller@comcast.com	Dan	Miller
9	dominiquelefebvre@gmail.com	Dominique	Lefebvre
10	edfrancis@yahoo.ca	Edward	Francis
11	eduardo@woodstock.com.br	Eduardo	Martins
12	ellie.sullivan@shaw.ca	Ellie	Sullivan

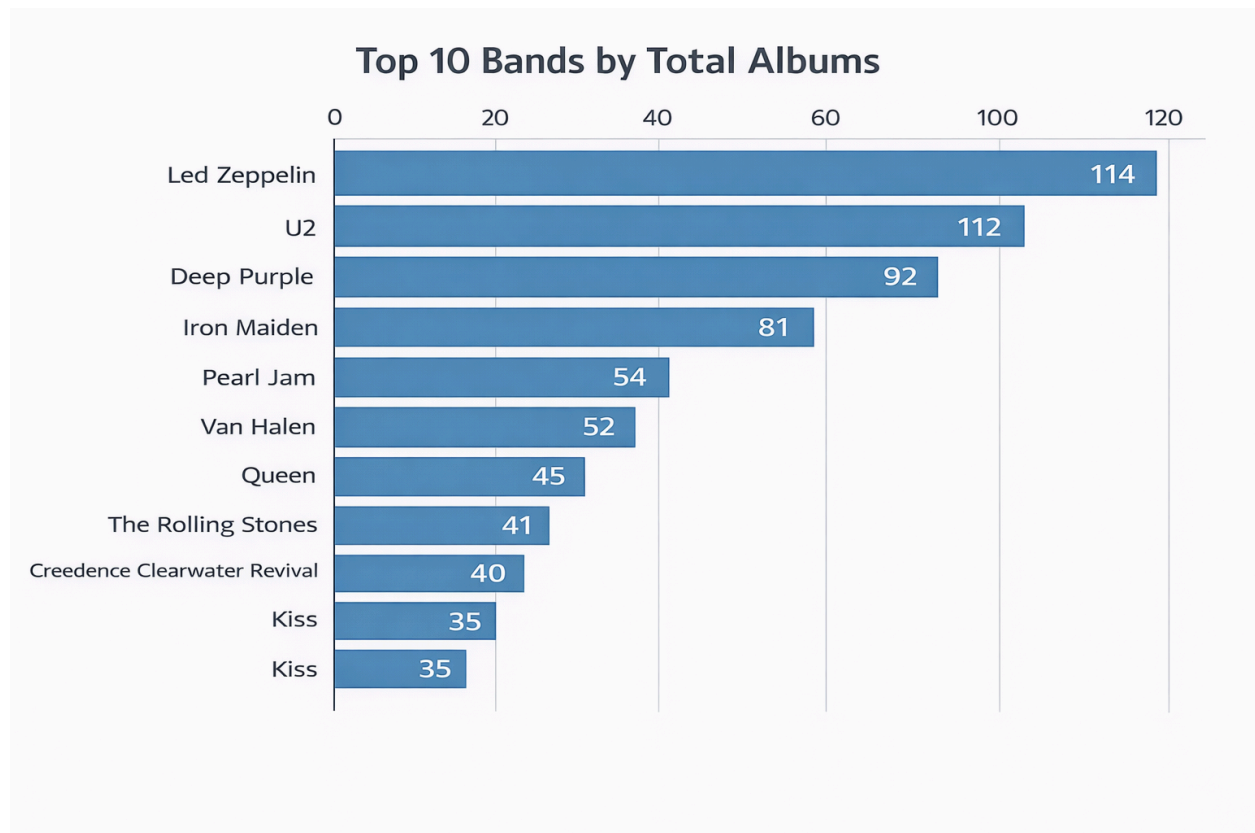
13	emma_jones@hotmail.com	Emma	Jones
14	enrique_munoz@yahoo.es	Enrique	Muñoz
15	fernadaramos4@uol.com.br	Fernanda	Ramos
16	fharris@google.com	Frank	Harris
17	fralston@gmail.com	Frank	Ralston
18	frantisekw@jetbrains.com	František	Wichterlová
19	ftremblay@gmail.com	François	Tremblay
20	fzimmermann@yahoo.de	Fynn	Zimmermann
21	hannah.schneider@yahoo.de	Hannah	Schneider
22	hholy@gmail.com	Helena	Holý
23	hleacock@gmail.com	Heather	Leacock
24	hughoreilly@apple.ie	Hugh	O'Reilly
25	isabelle_mercier@apple.fr	Isabelle	Mercier
26	jacksmith@microsoft.com	Jack	Smith
27	jenniferp@rogers.ca	Jennifer	Peterson
28	jfernandes@yahoo.pt	João	Fernandes

/*7.Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands*/

```

select a.name,
       count(*) as total_albums
from Musicstore_pro.artist as a
join Musicstore_pro.albums as al on a.artist_id = al.artist_id
join Musicstore_pro.track as t on al.album_id = t.album_id
where t.track_id in (
    select t.track_id
    from Musicstore_pro.track t
    join Musicstore_pro.genre g on t.genre_id = g.genre_id
    where g.name like "Rock"
)
group by a.name
order by total_albums desc
limit 10;

```



Key Insights

1. Clear Market Leaders

- **Led Zeppelin (114)** and **U2 (112)** dominate the list
- They are significantly ahead → strong catalog depth & legacy

2. Strong Second Tier

- **Deep Purple (92)** and **Iron Maiden (81)**
- These bands also have massive discographies but still **~20–30 albums behind leaders**

3. Mid-Level Performers

- Bands like **Pearl Jam (54)**, **Van Halen (52)**, and **Queen (45)**
- Solid output but almost **half of top performers**

4. Long Tail Distribution

- Remaining bands (**Rolling Stones, CCR, Kiss**) range between **35–41 albums**
- Shows a **gradual decline**, not a sharp drop

5. Insight for Business / Interview

You can say:

“The distribution shows a **top-heavy pattern**, where a few bands contribute disproportionately high album counts, followed by a steady decline across others.”

/* 8.Return all the track names that have a song length longer than the average song length. Return the Name and Milliseconds for each track. Order by the song length with the longest songs listed first. */

```
select name,milliseconds
from Musicstore_pro.track
where milliseconds > (select avg(milliseconds) from Musicstore_pro.track )
order by milliseconds desc;
```

/*9.Find how much amount spent by each customer on artists? Write a query to return customer name, artist name and total spent.*/

with best_sell_art as (

```

select a2.name,a2.artist_id,sum(l.unit_price * l.quantity) as total_sell
from Musicstore_pro.invoice_line l
join Musicstore_pro.track t on l.track_id = t.track_id
join Musicstore_pro.albums a on a.album_id = t.album_id
join Musicstore_pro.artist a2 on a2.artist_id = a.artist_id
group by a2.name,a2.artist_id
order by total_sell desc
limit 1
)
select c.customer_id,
       c.first_name,
       c.last_name,
       bsa.name,
       sum(l.quantity * l.unit_price) as amount_spent
from Musicstore_pro.customer c
join Musicstore_pro.invoice i on c.customer_id = i.customer_id
join Musicstore_pro.invoice_line l on l.invoice_id = i.invoice_id
join Musicstore_pro.track t on t.track_id = l.track_id
join Musicstore_pro.albums a on a.album_id = t.album_id
join best_sell_art as bsa on bsa.artist_id = a.artist_id
group by 1,2,3,4
order by amount_spent desc;

```

Rank	customer_id	Customer Name	amount_spent
1	46	Hugh O'Reilly	27.72
2	38	Niklas Schröder	18.81
3	3	François Tremblay	17.82
4	34	João Fernandes	16.83
5	53	Phil Hughes	11.88
6	41	Marc Dubois	11.88
7	47	Lucas Mancini	10.89
8	33	Ellie Sullivan	10.89
9	5	František Wichterlová	3.96

10 20 Dan Miller 3.96

(remaining customers spend between 0.99 – 2.97)

1. Top Heavy Spending

- **Hugh O'Reilly (27.72)** is a clear outlier
- Spent **~45% more than 2nd place**

2. Strong Top 4 Segment

- Top 4 customers contribute a **major portion of revenue**
- After that → sharp drop

3. Mid Tier Cluster

- Customers spending around **10–12**
- Represents **loyal but not premium buyers**

4. Long Tail

- Majority of customers spend only **0.99 – 2.97**
- Indicates:
 - Many **casual listeners**
 - Few **high-value customers**

5. Business Insight

“Revenue is highly concentrated among a small group of top customers, while the majority contribute minimal spend, indicating a classic long-tail distribution.”

/*10.We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared return all Genres.*/

```
WITH popular_genre AS
(
    SELECT COUNT(invoice_line.quantity) AS purchases, customer.country, genre.name,
    genre.genre_id,
    ROW_NUMBER() OVER(PARTITION BY customer.country ORDER BY
COUNT(invoice_line.quantity) DESC) AS RowNo
    FROM Musicstore_pro.invoice_line
    JOIN Musicstore_pro.invoice ON invoice.invoice_id = invoice_line.invoice_id
    JOIN Musicstore_pro.customer ON customer.customer_id = invoice.customer_id
    JOIN Musicstore_pro.track ON track.track_id = invoice_line.track_id
    JOIN Musicstore_pro.genre ON genre.genre_id = track.genre_id
    GROUP BY 2,3,4
    ORDER BY 2 ASC, 1 DESC
)
SELECT * FROM popular_genre WHERE RowNo <= 1;
```

Row	Country	Genre	Purchases
1	Argentina	Alternative & Punk	17
2	Australia	Rock	34
3	Austria	Rock	40
4	Belgium	Rock	26
5	Brazil	Rock	205
6	Canada	Rock	333
7	Chile	Rock	61

8	Czech Republic	Rock	143
9	Denmark	Rock	24
10	Finland	Rock	46
11	France	Rock	211
12	Germany	Rock	194
13	Hungary	Rock	44
14	India	Rock	102
15	Ireland	Rock	72
16	Italy	Rock	35
17	Netherlands	Rock	33
18	Norway	Rock	40
19	Poland	Rock	40
20	Portugal	Rock	108
21	Spain	Rock	46
22	Sweden	Rock	60
23	USA	Rock	561
24	United Kingdom	Rock	166

Key Insights:

1. Rock Dominates Globally

- Almost **every country's top genre = Rock**
- Only exception: **Argentina (Alternative & Punk)**

2. USA is a Massive Outlier

- **561 purchases** → far ahead of others
- Almost **2x Canada (333)**

“Rock is the dominant genre across almost all countries, with the USA contributing disproportionately high purchases, indicating a strong market concentration.”

/*11. Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount*/

```
with rk as (select c.customer_id,
                 c.first_name,
                 c.last_name,
                 c.country,
                 row_number()over(partition by c.country order by sum(i.total) desc ) as rn,
                 sum(i.total) as total_spend
from Musicstore_pro.customer c
join Musicstore_pro.invoice i on c.customer_id = i.customer_id
group by 1,2,3,4
order by total_spend desc)

select *
from rk
where rn <= 1
```

Rank	Customer Name	Country	Total Spend
1	František Wichterlová	Czech Republic	144.54
2	Hugh O'Reilly	Ireland	114.84
3	Manoj Pareek	India	111.87
4	Luís Gonçalves	Brazil	108.90
5	João Fernandes	Portugal	102.96
6	François Tremblay	Canada	99.99
7	Wyatt Girard	France	99.99

8	Jack Smith	USA	98.01
9	Enrique Muñoz	Spain	98.01
10	Phil Hughes	United Kingdom	98.01
11	Luis Rojas	Chile	97.02
12	Fynn Zimmermann	Germany	94.05
13	Mark Taylor	Australia	81.18
14	Terhi Hämäläinen	Finland	79.20
15	Ladislav Kovács	Hungary	78.21
16	Stanisław Wójcik	Poland	76.23
17	Joakim Johansson	Sweden	75.24
18	Bjørn Hansen	Norway	72.27
19	Astrid Gruber	Austria	69.30
20	Johannes Van der Berg	Netherlands	65.34
21	Daan Peeters	Belgium	60.39
22	Lucas Mancini	Italy	50.49
23	Diego Gutiérrez	Argentina	39.60
24	Kara Nielsen	Denmark	37.62

Key Insights

1. Top Spending Customer

- **František (Czech Republic)** leads with **144.54**
- Clear top outlier compared to others

2. Strong Top Tier (100+)

- Ireland, India, Brazil, Portugal
 👉 These countries have **high-value customers**

3. Balanced Middle Segment (~90–100)

- Canada, France, USA, UK, Spain
 👉 Stable and consistent markets

4. Gradual Decline Pattern

- Smooth drop from **100** → **37**
- No sudden crash → indicates **healthy distribution**

5. Lower Tier Markets

- Argentina, Denmark, Italy
 👉 Lower individual customer spending

Insight

“Top customers across countries show a gradual spending distribution with a few high-value outliers, indicating opportunities for targeted retention and upselling strategies.”

- Even smaller countries have **dedicated high spenders**
- Suggests:
 - Global reach is strong
 - Not dependent on a single region
- Segment customers into **high / medium / low value**
- Create a **Power BI dashboard story (this will impress recruiters a lot)**